Liam O'Connor

E-mail: liam.oconnor@u.northwestern.edu
Website: liamoconnor9.github.io

Education

Current **Ph.D., Applied Mathematics**, *Northwestern University* — Graduation December 2025.
Management for Ph.D.s certificate, *Kellogg School of Management, Northwestern University* — Summer 2025.

2020-2022 **M.S., Applied Mathematics**, *Northwestern University*.

2015-2019 **B.S., Applied Mathematics and Mechanical Engineering**, *University of Colorado Boulder* (double major, Cum laude).

Research Interests

1. **Astrophysical and Geophysical Fluid Dynamics:** magnetohydrodynamics, convection, particle tracking.
2. **Constrained Optimization:** control theory, inverse problems, nonlinear programming, quasi-Newton methods, adjoint methods.
3. **Data Analysis:** machine learning (neural networks), Monte Carlo methods, dimensionality reduction, statistical modelling.

Fellowships and Awards

1. **Graduate Research Fellowship Program**, *National Science Foundation* (2022-Current) — $147,000 personal research funding.
2. Walter P. Murphy Fellowship, *Northwestern University* (September 2020-September 2021).
3. Engineering Honors Residential Program, *University of Colorado Boulder* (August 2015-December 2019).
4. Engineering Excellence Fund Grant, *University of Colorado Boulder* (November 2018).
5. Scholar Athlete Award, *Collegiate Water Polo Association* (2017-2018).

Relevant Experience

2020-Current **Ph.D. Candidate**, *Center for Interdisciplinary Exploration and Research in Astrophysics (CIERA)*.

2018-2020 **Software Engineer**, *ANSYS*.

2017-2019 **Flight Development Intern**, *Laboratory for Atmospheric and Space Physics (LASP)*.

2015-2017 **Undergraduate Research Assistant**, *Center for Astrophysics and Space Astronomy (CASA)*.

Publications

1. Liam O'Connor, Daniel Lecoanet, Geoffrey M. Vasil, Kyle C. Augustson, Florentin Daniel, Evan H. Anders, Keaton J. Burns, Jeffrey S. Oishi, Benjamin P. Brown, "The Unsteady Taylor-Vortex Dynamo is Fast." **Physical Review Letters** (under review, 2025)
2. Liam O'Connor, Daniel Lecoanet, Evan H. Anders, Kyle C. Augustson, Keaton J. Burns, Geoffrey M. Vasil, Jeffery S. Oishi, Benjamin P. Brown, "Iterative Methods for Navier–Stokes Inverse Problems." accepted by **Physical Review E**, (2024). https://doi.org/10.1103/PhysRevE.109.045108
3. Christine Brennan, Mara Louise Smith, Rachael R Baiduc, Liam O'Connor, "Speech, Language, Hearing, and Otopathology Results From the International Smith-Magenis Syndrome Patient Registry." **Journal of Speech, Language, and Hearing Research**, (2024). https://doi.org/10.1044/2023_jslhr-23-00179
4. Liam O'Connor, Daniel Lecoanet, Evan H. Anders, "Marginally-Stable Thermal Equilibria of Rayleigh–Bénard Convection." **Physical Review Fluids**, (2021). https://dx.doi.org/10.1103/PhysRevFluids.6.093501

Patents

1. Liam O'Connor, Robert Schouten, Bruno Geoly, Niki Duer, Lucas McConnel, "Dynamic Canine Limb Prosthesis." **USPTO** Provisional Patent 62/841,149, (expired 2020).

Conference Proceedings

1. Liam O'Connor, Daniel Lecoanet, "Oscillatory Dynamos in Rotating Plane Couette Flow."
   77th Annual Meeting of the Division of Fluid Dynamics (DFD), APS (2024).

2. Liam O'Connor, Daniel Lecoanet, Evan H. Anders, Kyle C. Augustson, Keaton J. Burns, Geoffrey M. Vasil, Jeffery S. Oishi, Benjamin P. Brown, "Iterative Methods for Navier–Stokes Inverse Problems."
   76th Annual Meeting of the Division of Fluid Dynamics (DFD), APS (2023).

3. Liam O'Connor, Daniel Lecoanet, Evan H. Anders, Kyle C. Augustson, Keaton J. Burns, Geoffrey M. Vasil, Jeffery S. Oishi, Benjamin P. Brown, "Automated Adjoint-Looping Optimization with Pseudospectral Simulations."
   75th Annual Meeting of the Division of Fluid Dynamics (DFD), APS (2022).

4. Liam O'Connor, Daniel Lecoanet, Evan H. Anders, "Marginally-Stable Thermal Equilibria of Rayleigh–Bénard Convection."
   74th Annual Meeting of the Division of Fluid Dynamics (DFD), APS (2021).

Certificates

1. Research Communication Training Program (RCTP), *Northwestern University*

2. CU Business Intensive Certificate (CUBIC), *University of Colorado Boulder*

3. Certified Solidworks Professional (CSWP), *Dassault Systèmes*

Pedagogy, Outreach, and Community Service

Member, Chicago Juggling Club (Current).

Mentor, independent high school research project (2024).
In the convective regions of sun-like stars, opacity can have sensitive temperature-dependence due to the narrowband absorption of photons in species such as Iron. To investigate this phenomenon, a local high school student and I are performing convection simulations with nonlinear thermal diffusivity.

Academic Chair, Society of Industrial and Applied Mathematics (SIAM), Northwestern University Chapter (2024)
I organize the weekly applied math journal club at Northwestern University where we present original research on various subjects including fluid dynamics, cellular differentiation, and machine learning.

Mentor, Research Experiences in Astronomy at CIERA for High school students (REACH) (summer 2023)
Under my supervision, a local high school student performed convection simulations to measure the aspect ratio of convective cells.

Mentor, Research Experiences in Astronomy at CIERA for High school students (REACH) (summer 2022)
Under my supervision, a local high school student performed immersed boundary method simulations to measure the drag of projectiles with various geometries.

Vice-President, Society of Industrial and Applied Mathematics (SIAM), Northwestern University Chapter (2022-2023)
Organized department events and travel. We attended the Chicago Area SIAM Student Conference in 2022 and 2024.

Volunteer, MetroSquash afterschool program (2021-2022)
Assisted local middle school students in their routine homework assignments.

Technical Skills

Computer Programming
Bash, C, C++, C#, CUDA, Dedalus, Distributed Programming, FFTW, High Performance Computing (HPC), LabVIEW, LaPacke, Latex, Mathematica, MATLAB, Microsoft Office, MPI, Octave, OpenBLAS, OpenMP, Python, PyTorch, TensorFlow, Visual Studio.

Engineering Design and Analysis
Abaqus, ANSYS AIM, ANSYS DiscoveryLive, ANSYS Fluent, ANSYS Mechanical APDL, ANSYS SpaceClaim, ANSYS Workbench, AutoCAD, Computational Fluid Dynamics (CFD), Finite Element Analysis (Structural, Thermal, Vibrational, Optical), Geometric Dimensioning and Tolerancing (GDT), Solidworks.

Machine Fabrication
CNC (Mill, Lathe), Fused Deposition Modeling (FDM), Laser Cutter, Machine Maintenance, Plasma Cutter, Sand Casting, Stereolithography (SLA), Welding (MIG, TIG, Flux Core, Oxy-Acetylene), Wire EDM, Water Jet.